

Lecture 10: R-squared

Economics 326 — Introduction to Econometrics II

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Fitted values

- Consider the multiple regression model with k regressors: $Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i$.
- Let $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ be the OLS estimators.
- The **fitted (or predicted)** value of Y is: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \hat{\beta}_2 X_{2,i} + \dots + \hat{\beta}_k X_{k,i}$.
- The **residual** is: $\hat{U}_i = Y_i - \hat{Y}_i$.
- Consider the average of $\hat{Y}$:

$$\begin{aligned}\bar{\hat{Y}} &= \frac{1}{n} \sum_{i=1}^n \hat{Y}_i \\ &= \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{U}_i) \\ &= \bar{Y} - \frac{1}{n} \sum_{i=1}^n \hat{U}_i = \bar{Y},\end{aligned}$$

because **when there is an intercept**, $\sum_{i=1}^n \hat{U}_i = 0$.

Sum-of-Squares

- The **total** variation of Y in the **sample** is:

$$SST = \sum_{i=1}^n (Y_i - \bar{Y})^2 \text{ (Total Sum-of-Squares).}$$

- The **explained** variation of Y in the **sample** is:

$$SSE = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \text{ (Explained or Model Sum-of-Squares).}$$

- The **residual** (unexplained or error) variation of Y in the **sample** is:

$$SSR = \sum_{i=1}^n \hat{U}_i^2 \text{ (Residual Sum-of-Squares).}$$

- If the regression contains an intercept:

$$SST = SSE + SSR.$$

Proof of SST=SSE+SSR

- First,

$$\begin{aligned}
 SST &= \sum_{i=1}^n (Y_i - \bar{Y})^2 \\
 &= \sum_{i=1}^n (\hat{Y}_i + \hat{U}_i - \bar{Y})^2 \\
 &= \sum_{i=1}^n ((\hat{Y}_i - \bar{Y}) + \hat{U}_i)^2 \\
 &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n \hat{U}_i^2 + 2 \sum_{i=1}^n (\hat{Y}_i - \bar{Y}) \hat{U}_i \\
 &= SSE + SSR + 2 \sum_{i=1}^n (\hat{Y}_i - \bar{Y}) \hat{U}_i.
 \end{aligned}$$

- Next, we will show that $\sum_{i=1}^n (\hat{Y}_i - \bar{Y}) \hat{U}_i = 0$.

Proof of SST=SSE+SSR

- Since $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \dots + \hat{\beta}_k X_{k,i}$,

$$\begin{aligned}
 &\sum_{i=1}^n (\hat{Y}_i - \bar{Y}) \hat{U}_i \\
 &= \sum_{i=1}^n ((\hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \dots + \hat{\beta}_k X_{k,i}) - \bar{Y}) \hat{U}_i \\
 &= \hat{\beta}_0 \sum_{i=1}^n \hat{U}_i + \hat{\beta}_1 \sum_{i=1}^n X_{1,i} \hat{U}_i + \dots + \hat{\beta}_k \sum_{i=1}^n X_{k,i} \hat{U}_i - \bar{Y} \sum_{i=1}^n \hat{U}_i.
 \end{aligned}$$

- The OLS normal equations for a model with an intercept:

$$\sum_{i=1}^n \hat{U}_i = \sum_{i=1}^n X_{1,i} \hat{U}_i = \dots = \sum_{i=1}^n X_{k,i} \hat{U}_i = 0.$$

- It follows that $\sum_{i=1}^n (\hat{Y}_i - \bar{Y}) \hat{U}_i = 0$.

R^2

- Consider the following measure of goodness of fit:

$$\begin{aligned}
 R^2 &= \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \\
 &= \frac{SSE}{SST} \\
 &= 1 - \frac{SSR}{SST} \\
 &= 1 - \frac{\sum_{i=1}^n \hat{U}_i^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}.
 \end{aligned}$$

- $0 \leq R^2 \leq 1$.
- R^2 measures the proportion of variation in Y **in the sample** explained by the X 's.

SSR is non-increasing in regressors

- Consider two models:

$$Y_i = \tilde{\beta}_0 + \tilde{\beta}_1 X_{1,i} + \tilde{U}_i,$$

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \hat{\beta}_2 X_{2,i} + \hat{U}_i.$$

- **Claim:** Adding a regressor cannot increase SSR :

$$\sum_{i=1}^n \tilde{U}_i^2 \geq \sum_{i=1}^n \hat{U}_i^2.$$

- This generalizes to the case of k and $k+1$ regressors.

Proof: constrained vs unconstrained

- By definition, OLS minimizes the sum of squared residuals.
- The **short** regression solves

$$\min_{b_0, b_1} \sum_{i=1}^n (Y_i - b_0 - b_1 X_{1,i})^2,$$

which is the same as minimizing $\sum_{i=1}^n (Y_i - b_0 - b_1 X_{1,i} - b_2 X_{2,i})^2$ subject to the **constraint** $b_2 = 0$.

- The **long** regression minimizes the same sum of squares **without** the constraint $b_2 = 0$.
- The minimum over a **larger** set (unconstrained) cannot exceed the minimum over a **smaller** set (constrained):

$$\underbrace{\sum_{i=1}^n \hat{U}_i^2}_{\text{unconstrained min}} \leq \underbrace{\sum_{i=1}^n \tilde{U}_i^2}_{\text{constrained min}}.$$

R^2 is non-decreasing in regressors

- Since $SST = \sum_{i=1}^n (Y_i - \bar{Y})^2$ does not depend on the regressors, and SSR cannot increase when regressors are added:

$$R^2 = 1 - \frac{SSR}{SST}$$

cannot decrease when more regressors are added, **even if the additional regressors are irrelevant**.

Adjusted R^2

- Since R^2 cannot decrease when more regressors are added, **even if the additional regressors are irrelevant**, an alternative measure of goodness-of-fit has been developed.
- **Adjusted R^2** : the idea is to adjust SSR and SST for degrees of freedom:

$$\bar{R}^2 = 1 - \frac{SSR/(n-k-1)}{SST/(n-1)}.$$

- $\bar{R}^2 < R^2$.
- $\bar{R}^2$ can decrease when more regressors are added.

Estimation of σ^2

- In the multiple linear regression model, we can estimate $\sigma^2 = E[U_i^2 | \mathbf{X}]$ as follows:

Let

$$\hat{U}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1,i} - \hat{\beta}_2 X_{2,i} - \dots - \hat{\beta}_k X_{k,i}.$$

- An estimator for σ^2 is

$$\begin{aligned} s^2 &= \frac{1}{n-k-1} \sum_{i=1}^n \hat{U}_i^2 \\ &= \frac{SSR}{n-k-1}. \end{aligned}$$

- The adjustment $k+1$ is for the number of parameters we have to estimate in order to construct the $\hat{U}$'s:

$$\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k.$$

Unbiasedness of s^2

$$s^2 = \frac{1}{n-k-1} \sum_{i=1}^n \hat{U}_i^2.$$

- s^2 is an unbiased estimator of σ^2 (i.e., $E[s^2 | \mathbf{X}] = \sigma^2$) if the following conditions hold:
 1. $Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i$.
 2. Conditional on $\mathbf{X}$, $E[U_i | \mathbf{X}] = 0$ for all i 's.
 3. Conditional on $\mathbf{X}$, $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all i 's (homoskedasticity).
 4. Conditional on $\mathbf{X}$, $E[U_i U_j | \mathbf{X}] = 0$ for all $i \neq j$.

R example

- Using the `hprice1` dataset from the `wooldridge` package, we regress house price on square footage, number of bedrooms, and lot size ($n = 88$, $k = 3$):

```
library(wooldridge)
m <- lm(price ~ sqft + bdrms + lotsize, data = hprice1)
summary(m)
```

- The `summary()` output reports:

Residual standard error: 59.83 on 84 degrees of freedom

Multiple R-squared: 0.6724, Adjusted R-squared: 0.6607

- From here we can read off $R^2 = 0.6724$, $\bar{R}^2 = 0.6607$, and $s = 59.83$.
- The **residual degrees of freedom** is $n - k - 1 = 88 - 3 - 1 = 84$.

Recovering SSR, SST, SSE from R output

- Since $s = 59.83$, we have $s^2 = 59.83^2 \approx 3,580$.
- $SSR = s^2 \cdot (n - k - 1) \approx 3,580 \times 84 \approx 300,720$.
- From $R^2 = 1 - SSR/SST$:

$$SST = \frac{SSR}{1 - R^2} \approx \frac{300,720}{1 - 0.6724} = \frac{300,720}{0.3276} \approx 918,100.$$

- $SSE = SST - SSR \approx 918,100 - 300,720 = 617,380$.